LiFE Companion: A Lightweight Citizen Engagement Platform for Mission LiFE

HackCult '26 · Track HC26-GEDA01 (Green Energy & Sustainability)

Abstract

Awareness of climate change is high but daily habits change slowly. We present LiFE Companion, a mobile-first web application that connects personal footprint measurement, contextual energy nudges, gamified community action and a rooftop-solar feasibility tool. The prototype is a zero-dependency single-page app with an optional FastAPI backend. We describe design, models, assumptions and limitations. No user study has been conducted; effectiveness claims are hypotheses.

1. Problem

Citizens lack personalised tools to measure footprints, receive real-time reduction advice and see the effect of their actions (GEDA problem statement). The gap between national goals and household behaviour is the target.

2. Related ideas

Behavioural-science work on feedback, social comparison and commitment suggests that frequent, specific feedback and peer norms can reduce household energy use. We adopt these as design principles (evidence-supported in literature; effect size in our setting is unknown).

3. System design

Client-side modules compute everything offline for low-connectivity use. State persists in localStorage. The backend exposes /footprint, /points and /leaderboard for multi-user deployment, backed by PostgreSQL or Firestore.

Component	Method	Basis
Footprint	Sum of activity x emission factor	Grid 0.82 kg/kWh; LPG 2.98 kg/kg; petrol 2.31 kg/L
Nudges	Time-of-day rules	Evening peak 18-22h; solar midday
Gamification	10 pts/task, leaderboard	Behavioural reinforcement
Solar	Yield x tariff, subsidy-adjusted	1500 kWh/kW/yr; Rs 55k/kW

4. Footprint model

CO_2e = sum(activity x factor). Factors are editable assumptions drawn from approximate Indian grid and fuel values; they must be verified against current CEA/IPCC sources before official use.

5. Nudge engine

Rules map hour-of-day to advice: shift heavy loads out of the 6-10 pm peak, use midday solar, charge off-peak. Future work replaces rules with grid-signal and weather APIs.

6. Gamification

Each micro-task awards points and logged CO_2 savings, which feed a community goal bar and leaderboard. Self-reporting is unverified; anti-cheat is future work.

7. Solar feasibility

Capacity = min(roof area/10 m^2, annual demand/yield). Savings = generation x tariff; payback = net cost/savings after subsidy. Assumptions are approximate and flagged in the UI.

8. Scalability and security

Stateless API, horizontal scaling, indexed leaderboard queries, caching, JWT auth and rate limiting are planned. The prototype has no authentication.

9. Limitations

Simulated leaderboard; unvalidated factors; no user testing; no live grid data; self-reported actions.

10. Roadmap

Auth, cloud sync, OpenWeather irradiance, Maps vendor finder, regional languages, smart-meter integration, pilot study with measured consumption.

11. Conclusion

LiFE Companion shows that a small, offline-capable app can cover the four GEDA core features. Real impact requires a pilot with measured data.